

S#	Student Name	Edu Email	Student ID	Marks		
				Report & Slides (30)	Implementation (50)	Presentation (20)
1	Abdulrahman Ghareeb AbdulSattar Muhammad	abdelrahman.ghareeb21@feng.bu.edu.eg	221902911			
2	Ahmed Hamdi AbdulSattar Jadallah	ahmed403176@feng.bu.edu.eg	221903084			
3	Hisham Ahmed Mohamed Abaza	hesham.ahmed21@feng.bu.edu.eg	221902963			
4	Abdulaziz Ahmed Abdulaziz Abdalnabi	abdulaziz414046@feng.bu.edu.eg	221903195			
5	Muhannad Mahmoud AbdulWahab Ahmed	mohannad414226@feng.bu.edu.eg	221903201			
6	Ahmed Amr Ahmed Hashem	ahmed412933@feng.bu.edu.eg	231903714			
7	Rawan Abdulaziz Ali Abdulaziz	rawan403225@feng.bu.edu.eg	221903115			
8	Ahmed Mahmoud Mohamed Abdullatif	ahmed195140@feng.bu.edu.eg	211902086			

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Report & Slides (35)		Points
1. Content	Project includes all material needed to give a good understanding of the topic.	10
	All content throughout the paper & presentation is accurate. There are no factual errors.	10
2. Format	Sequencing of Information: Information is organized in a clear, logical way. It is easy to anticipate the next slide.	5
	All graphics are attractive (size and colors) and support the topic of the presentation.	5
	Text-Font Choice & Formatting: Font formats (color, bold, italic) have been carefully planned to enhance readability and content.	5
Presentation (20)		
1. Delivery	Members understand presented material and can clearly answer all questions about it.	10
	Members spoke at a good rate, volume and with good grammar. They maintained eye-contact while using, but not reading their notes.	5
2. Cooperation	Group's members share tasks and all performed responsibly all of the time.	5
Implementation (45)		
1. Code	Members understand the code and can clearly answer all questions about it.	20
	The code is completely functional and correctly producing the outputs.	25
Total Score		100

Discrete Random Variables

1. Bernoulli Discrete Distribution

a) Unveiling the essence of Bernoulli's Distribution:

The Bernoulli distribution is relevant to situations involving a single trial with two potential outcomes, commonly referred to as Bernoulli trials. Imagine any experiment posing a binary question such as whether a coin will land on heads, if a die will roll a six, if an ace will be picked from a deck of cards, or if voter X will vote "yes" in a political referendum, among others.

In the realm of Bernoulli trials, these two outcomes can be conceptualized as "success" or "failure," but it's important to note that these terms don't carry a literal interpretation. In this context, "success" simply denotes achieving a positive result, such as rolling a six or picking an ace.

The essence of the Bernoulli distribution lies in its ability to model the potential outcomes of a Bernoulli trial. Essentially, it provides a framework for calculating the probabilities associated with each of the two outcomes whenever an event entails only two possible results.

To succinctly encapsulate its essence within the confines of mathematical expressions, we can articulate the concept as follows:

As elucidated earlier, given its binary nature, denoted by the variable X representing a random variable conforming to the Bernoulli's distribution, the probability mass function of X elegantly manifests as either 0 or 1.

* LET $X = 1$ if a success occurs, and $X = 0$ if a failure occurs.

$$P(X = x) = p^x(1 - p)^{1-x}$$

$$\text{for } x = 1, P(X = 1) = p^1(1 - p)^0 = p$$

$$\text{for } x = 0, P(X = 0) = p^0(1 - p)^1 = 1 - p$$

And that's totally comprehensible, since both are complements of each other.

b) As for illustrating this concept through the graphical representation of its PMF:

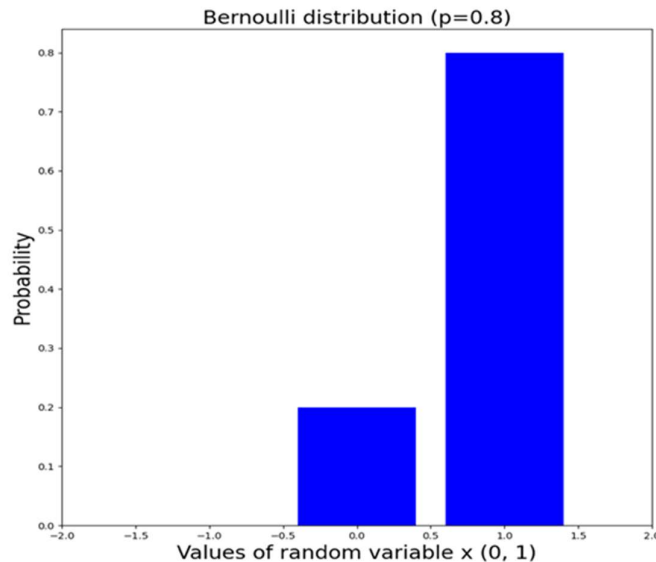


Fig. An instance of Bernoulli's distribution with a parameter of $p=0.8$

c) Rademacher Random Variable: An Overview:

The Rademacher random variable is a discrete random variable that takes values in the set $\{-1, 1\}$. The probability mass function (PMF) of the Rademacher random variable is defined as follows:

$$P(X = 1) = P(X = -1) = \frac{1}{2}$$

This implies that the Rademacher random variable is equally likely to take on the values of 1 or -1 with equal probability. This also implies that it is symmetrical around zero. This means that the probability of obtaining a positive value is the same as the probability of obtaining a negative value. Another property of RRV is that it can maximize the variance of Bernoulli's random variable.

2. Binomial Discrete Distribution

a) Introduction:

To start, the "binomial" in binomial distribution means two terms. We're interested not just in the number of successes, nor just the number of attempts, but in both. Each is useless to us without the other. Binomial distribution is a common discrete distribution used in statistics, as opposed to a continuous distribution, such as normal distribution. This is because binomial distribution only counts two states, typically represented as 1 (for a success) or 0 (for a failure) given a number of trials in the data. Binomial distribution thus represents the probability for x successes in n trials, given a success probability p for each trial. Binomial

distribution summarizes the number of trials, or observations when each trial has the same probability of attaining one particular value. Binomial distribution determines the probability of observing a specified number of successful outcomes in a specified number of trials.¹ Binomial distribution is often used in social science statistics as a building block for models for dichotomous outcome variables, such as whether a Republican or Democrat will win an upcoming election, whether an individual will die within a specified period of time, etc. It also has applications in finance, banking, and insurance, among other industries. Analyzing Binomial Distribution The expected value, or mean, of a binomial distribution is calculated by multiplying the number of trials (n) by the probability of successes (p), or $n \times p$. For example, the expected value of the number of heads in 100 trials of heads or tails is 50, or (100×0.5) . Another common example of binomial distribution is by estimating the chances of success for a free-throw shooter in basketball, where 1 = a basket made and 0 = a miss. The binomial distribution formula is calculated as:

$$P(x;n,p) = nCx \times p^x (1-p)^{n-x}$$

where:

n is the number of trials (occurrences)

x is the number of successful trials

p is probability of success in a single trial

nCx is the combination of n and x .

A combination is the number of ways to choose a sample of x elements from a set of n distinct objects where order does not matter, and replacements are not allowed. Note that $nCx = n! / (x!(n-x)!)$, where $!$ is factorial (so, $4! = 4 \times 3 \times 2 \times 1$). The mean of the binomial distribution is np , and the variance of the binomial distribution is $np(1-p)$. When $p = 0.5$, the distribution is symmetric around the mean. When $p > 0.5$, the distribution is skewed to the left. When $p < 0.5$, the distribution is skewed to the right. The binomial distribution is the sum of a series of multiple independent and identically distributed Bernoulli trials. In a Bernoulli trial, the experiment is said to be random and can only have two possible outcomes: success or failure. For instance, flipping a coin is considered to be a Bernoulli trial; each trial can only take one of two values (heads or tails), each success has the same probability (the probability of flipping a head is 0.5), and the results of one trial do not influence the results of another.² Bernoulli distribution is a special case of binomial distribution where the number of trials $n = 1$. Example of Binomial Distribution Binomial distribution is calculated by multiplying the probability of success raised to the power of the number of successes and the probability of failure raised to the power of the difference between the number of successes and the number of trials. Then, multiply the product by the combination between the number of trials and the number of successes. For example, assume that a casino created a new game in which participants are able to place bets on the number of heads or tails in a specified number of coin flips. Assume a participant wants to place a \$10 bet that there will be exactly six heads in 20-coin flips. The participant wants to calculate the probability of this occurring, and therefore, they use the calculation for binomial distribution. The probability was calculated as $(20! / (6! \times (20 - 6)!)) \times (0.50)^6 \times (1 - 0.50)^{(20 - 6)}$. Consequently, the probability of exactly six heads occurring in 20-coin flips is 0.037, or 3.7%. The expected value was 10 heads in this case, so the participant made a poor bet.

So how can this be used in finance? One example: Let's say you're a bank, a lender, that wants to know within three decimal places the likelihood of a particular borrower defaulting. What are the chances of so many borrowers defaulting that they would render the bank

insolvent? Once you use the binomial distribution function to calculate that number, you have a better idea of how to price insurance, and ultimately how much money to lend out and how much to keep in reserve.

b) What is binomial distribution?

Binomial distribution is a probability distribution used in statistics that states the likelihood that a value will take one of two independent values under a given set of parameters or assumptions.

c) How is binomial distribution used?

This distribution pattern is used in statistics but has implications in finance and other fields. Banks may use it to estimate the likelihood of a particular borrower defaulting or how much money to lend and the amount to keep in reserve. It's also used in the insurance industry to determine policy pricing and to assess risk.

d) Why is binomial distribution important?

Binomial distribution is used to figure the likelihood of a pass or fail outcome in a survey or experiment replicated numerous times. There are only two potential outcomes for this type of distribution. More broadly, distribution is an important part of analyzing data sets to estimate all the potential outcomes of the data and how frequently they occur. Forecasting and understanding the success or failure of outcomes is essential to business development.

3. Geometric Discrete Distribution

a) What is Geometric Distribution?

The geometric distribution is a discrete probability distribution that calculates the probability of the first success occurring during a specific trial. In other words, during a series of attempts, what is the probability of success first occurring during each attempt? Use this distribution when you need to understand how many attempts are necessary to produce the first successful outcome.

For example, the geometric distribution can answer the following questions. What is the probability of the first:

- Six in a series of die rolls?
- Year of catching the flu over the years?
- Person to support a law during a repeated sampling for an interview?
- Product to have a defect in a random sample from an assembly line?
- Successful attempt for a project or task?

This distribution is an example of a Probability Mass Function (PMF) because it calculates likelihoods for discrete random variables.

In this post, learn when to use the geometric distribution and its cumulative form, about its formula, and how to calculate probabilities by hand. I also include a geometric distribution calculator that you can use with what you learn.

b) The Geometric Distribution is Memoryless

The geometric distribution is “memoryless.” Memoryless is a distribution attribute indicating that the occurrence of the next success does not depend on when the last success occurred or when you start looking for successes. This type of process has independent events that occur with a constant probability.

For example, when you start observing die rolls looking for the next six, the probability of when the next six appears does not depend on when the last six appeared or when you start watching for sixes.

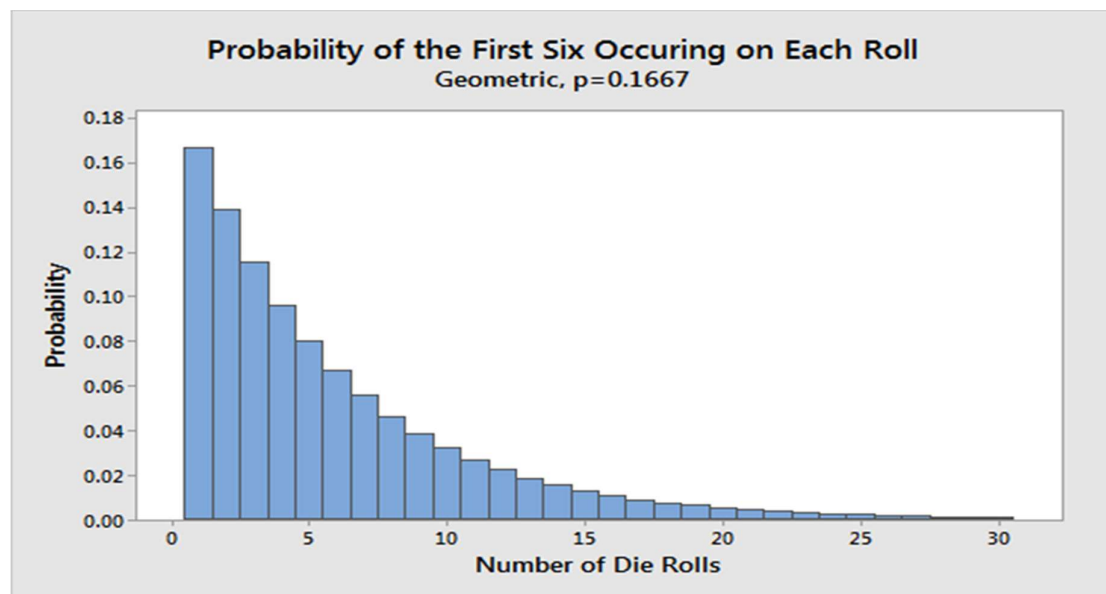
Both the geometric distribution and exponential distribution are memoryless.

c) Using the Geometric Distribution

The geometric distribution models the probabilities for the first event occurring during various trials when the likelihood of an event is known. Let’s bring it to life with an example!

Imagine you’re playing a game where rolling sixes on a die is beneficial. You’d like to know the probability of when you will throw the first six. Suppose you want to learn the chance of getting the first six on precisely the third roll. In this example, the probability (p) is $1/6 = 0.1667$, and we’re interested in the third roll.

The geometric distribution calculates that the probability of the first six occurring on the third die roll is 0.115755.



d) Geometric Cumulative Distribution Function

The geometric distribution is superb for understanding when an event might first occur.

However, many times you're not interested in a specific trial. For example, you might be interested in the probability of getting your first six within the first six rolls. In that case, you're interested in the total probability across multiple trials.

Technically, the geometric cumulative probability calculates the likelihood of obtaining the first event in less than or equal to N trials. If you need a $\geq$ probability, use the inverse geometric cumulative distribution. These days, most statistical software lets you specify the direction.

e) Geometric Distribution Assumptions and Notation

The geometric distribution models the probabilities for the first success occurring on the X^{th} trial. However, your data must meet the following requirements for the geometric distribution to be appropriate.

- **Your data must be binary:** For example, infected or uninfected, 6 or not 6, pass or fail, etc.
- **Independent trials:** One trial's result does not affect the next trial. For example, a coin toss doesn't affect the following coin toss.
- **The probability remains constant over time.** In some contexts, this supposition is true due to the physical attributes of the process, such as coin tosses and die rolls. However, the likelihood won't necessarily remain constant in other contexts. For example, the probability of a product defect at a manufacturing plant can change over time. Use a P chart (a control chart) to verify this assumption when the chance can change.

f) Geometric Experiments and Bernoulli Trials

Typically, you'll use the geometric distribution when you have Bernoulli Trials. These trials satisfy the binomial distribution assumptions above. In these trials, analysts label one outcome a success and the other a failure.

A Bernoulli trial contains attempts where the probability of success is constant. Geometric experiments perform a series of attempts until the first success.

g) Two Forms of the Geometric Distribution

Note that there are two forms of geometric distribution. The two forms model:

- The first success occurring on the X^{th} trial.
- The number of failures before the first success.

Fortunately, these two forms are equivalent. The difference comes down to how you count the trials. The former definition counts all trials, including the final, successful one. The latter definition counts only the failures before success. Success on the

X^{th} trial is equivalent to having $X - 1$ failures. Statisticians sometimes refer to the first form as the shifted geometric distribution because it moves the final trial over by counting the success.

For the die example, I calculated the probability of first rolling a six on the third trial. Using the number of failures form, we don't count the success trial, just the preceding failures. Hence, if we have success on the third roll, that means we have two failures, and those two forms of the geometric distribution calculate the same probability.

I use the success on the X^{th} trial version throughout this post, except for the geometric distribution calculator section, where we need to convert to the number of failures form.

h) Geometric Distribution Formula

Generally, you'll use statistical software or online calculators to calculate probabilities for the geometric distribution. However, I'll teach you how to use the geometric distribution formula so you can calculate them manually. Additionally, I'll walk you through the formula with a worked example.

The geometric distribution formula for the probability of the first success occurring on the X^{th} trial is the following:

$$P(x) = (1 - p)^{x-1}p$$

where:

- x is the number of trials.
- p is the probability of success for each trial.

The geometric distribution formula takes the probability of failure ($1 - p$) and raises it by the number of failures ($x - 1$). That produces the likelihood of having failures for all trials before the trial of interest (x). Then the equation multiplies the probability of failure by the probability of success (p) occurring on the trial of interest. That gives you the probability of the first success happening on the X^{th} trial.

The mean of a geometric distribution is $1 / P$. This mean is the expected value for a geometric distribution.

i) Worked Example of Finding a Geometric Distribution Probability

We'll use the geometric distribution formula to calculate the probability of rolling the first six on the third roll. That's the example we used before, and now we'll calculate it by hand. Here are the values to enter into the formula:

- $x = 3$
- $p = 0.1667$

$$\begin{aligned}
 P(x) &= (1 - p)^{x-1}p \\
 &= 0.8333^2 * 0.1667 \\
 &= 0.6944 * 0.1667 = 0.1158
 \end{aligned}$$

This calculation by hand confirms the previous statistical software results within rounding error.

If you need to calculate the cumulative geometric probability for a range of trials, calculate the probability for each attempt and then sum the probabilities.

For example, if you need to calculate the probability of the first six occurring in the first six rolls, calculate the likelihood of it happening on rolls 1, 2, 3, 4, 5, and 6. Then sum those probabilities.

Finally, the solution for the problem in the geometric distribution calculator section is that the cumulative probability of completing the project successfully during attempts 1 – 4 is 0.7599. To solve this problem:

1. Enter 0.3 for the Probability of success.
2. In Number of failures, enter 0, 1, 2, and 3 individually and record each probability.
3. Sum those probabilities

j) Real-life applications:

As the preceding example shows, a common application of the geometric distribution is in games of chance with two outcomes (a rolled die is either a 3 or not a 3, and a coin toss is either heads or tails). The geometric distribution is also used in sports—using past averages such as a pitcher’s strikeout rate in baseball or a pass completion rate in football (soccer)—to determine probabilities of certain events in games.

Geometric distribution also has applications in manufacturing, where it is important to monitor and identify defective products. If the defect rate of a production process is known, it can be used to determine the probability of finding a defective product during the quality-control process. Identifying this probability, in turn, helps to determine (and, ideally, improve) quality control.

In business and finance, geometric distributions can be used in a cost-benefit analysis, which is intended to aid managers’ decision making. This application, however, can be more challenging than the others described here, because the probabilities associated with individual events are subject to multiple internal and external factors that do not necessarily stay constant over time.

4. Poisson Discrete Distribution

a) What is Poisson distribution?

A **Poisson distribution** is a discrete probability distribution. It gives the probability of an event happening a certain number of times within a given interval of time or space and it has one parameter λ (lambda).

b) When is Poisson distribution used?

We use Poisson when we want to compute events that occur randomly at given time or distance e.g. number of phone calls received by cell tower or number of photons that hit photon detector or find number of customers arriving to store or car passing on the road in a given mile.

c) How is Poisson distribution used?

We need to find the probability of the events in a given interval, so we find it through the PMF of X , where X is Poisson random variable.

$$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{Where } k = 1, 2, \dots$$

To find the probability of random variable X that is less than or equal k we use the CDF function of Poisson:

$$F_X(k) = P[X \leq k] = \sum_{l=0}^k \frac{\lambda^l}{l!} e^{-\lambda}$$

- The Expectation/mean of Poisson:

It is the average of a set of values and equals,

$$E[X] = \lambda$$

- The variance of Poisson:

It is a measure of how far a set of data are spread out from there mean.

$$\text{Var}[X] = \lambda$$

d) What are the conditions of Poisson distribution?

Example (1):

The number of orders arriving per day for a small business is given by Poisson, and (λ) is the average number of orders arriving per day let $\lambda = 5$.

The PMF for this example is:

So, to draw the PMF we need to substitute k with numbers starting from 1 and get the probability at this state.

$$p_x(k) = \frac{5^k}{k!} e^{-5} \quad \text{where } k = 1, 2, 3, \dots$$

The CDF for this example is:

After that we are summing each state's probability with the previous probabilities to get the CDF of each state

$$F_x(k) = P[X \leq k] = \sum_{l=0}^k \frac{5^l}{l!} e^{-5}$$

The variance and expectation are:

Variance and mean are easy found because we already know lambda.

$$E[X] = \text{Var}[X] = 5$$

e) Poisson approximation to Binomial

In binomial when P is very small and n is very high we choose Poisson over binomial for simplicity.

In this case PMF of binomial equals PMF of Poisson

$$\frac{\lambda^k}{k!} e^{-\lambda} = \binom{n}{k} x^k a^{n-k}$$

Example (2):

A binary communication channel has probability of bit error of $p = 10^{-6}$. Suppose that transmission occurs in blocks of 10,000 bits.

The PMF for this example is to be get from Poisson for simplicity as n is large and p is small.

In this example we can get $\lambda = 10000 * 10^{-6} = 0.01$

$$p_x(k) = \frac{0.01^k}{k!} e^{-0.01}$$

The Expectation and Variance are equal and equal lambda = 0.01

So to draw the PMF we need to substitute k with a range of numbers starting from 1 and get the probability at this state and when plotting the PMF we plot the binomial and Poisson made sure of the equality of PMFs between binomial and Poisson and the results as shown in the figure they are having exactly the same curve.

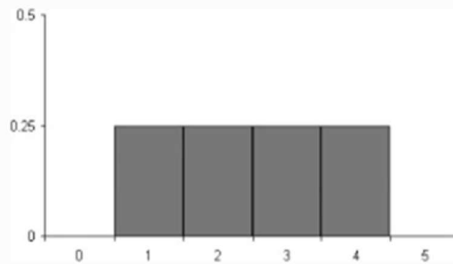
After that we are summing each state's probability with the previous probabilities to get the CDF of each state

5. Uniform discrete distribution

The discrete uniform distribution is a discrete probability distribution. The corresponding continuous probability distribution is the (continuous) uniform distribution. Consider n events, each of which have the same probability.

$P(X=x) = \frac{1}{n}$ the random variable X follows a discrete uniform distribution, and its probability function is:

$$P(X = x) = \begin{cases} \frac{1}{n}, & \text{if } x = x_1, x_2, \dots, x_n \\ 0, & \text{if not} \end{cases}.$$



Discrete uniform distribution, $n = 4$

a) What are some of its properties?

The **expected value** of the discrete uniform distribution over the set of first n natural numbers is, by definition, given by:

$$\begin{aligned} E[X] &= \sum_{x=1}^n x \cdot P(X = x) \\ &= \frac{1}{n} \cdot \sum_{x=1}^n x \\ &= \frac{1}{n} \cdot (1 + 2 + \dots + n) \\ &= \frac{1}{n} \cdot \frac{n^2 + n}{2} \\ &= \frac{n + 1}{2}. \end{aligned}$$

The **variance** of this discrete uniform distribution is equal to:

$$\text{Var}(X) = E[X^2] - (E[X])^2 .$$

Since

$$\begin{aligned} E[X^2] &= \sum_{x=1}^n x^2 \cdot P(X = x) \\ &= \frac{1}{n} \cdot \sum_{x=1}^n x^2 \\ &= \frac{1}{n} \cdot (1^2 + 2^2 + \dots + n^2) \\ &= \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} \\ &= \frac{2n^2 + 3n + 1}{6} , \end{aligned}$$

And

$$(E[X])^2 = \frac{(n+1)^2}{2^2}$$

We have:

$$\begin{aligned} \text{Var}(X) &= \frac{2n^2 + 3n + 1}{6} - \frac{(n+1)^2}{2^2} \\ &= \frac{n^2 - 1}{12} . \end{aligned}$$

The discrete uniform distribution is often used to generate random numbers from any discrete or continuous probability distribution.

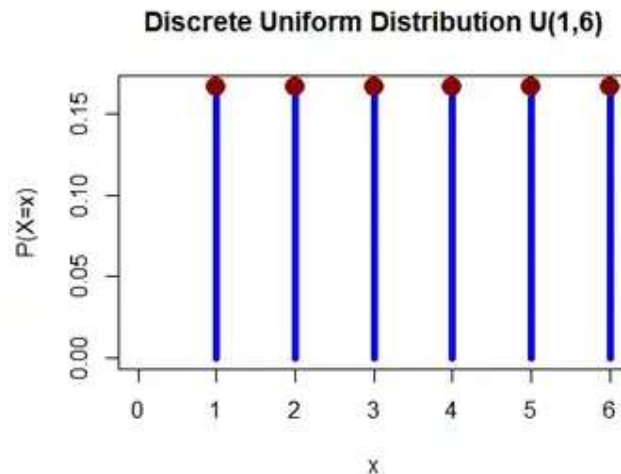
Consider the random variable X , the score obtained by throwing a die. If the die is not loaded, the probability of obtaining any particular score is equal to $\frac{1}{6}$. Therefore, we have:

$$\begin{aligned} P(X = 1) &= \frac{1}{6} \\ P(X = 2) &= \frac{1}{6} \\ &\dots \\ P(X = 6) &= \frac{1}{6} . \end{aligned}$$

The number of possible events is $n = 6$. We therefore have:

$$P(X = x) = \begin{cases} \frac{1}{6}, & \text{for } x = 1, 2, \dots, 6 \\ 0, & \text{if not} \end{cases}.$$

In other words, the random variable X follows the discrete uniform distribution.



b) How we can benefit from Discrete Uniform Distribution:

- **Random Number Generation:** Discrete uniform distributions are utilized in generating random integers within a specified range. For instance, in simulations, cryptography, or gaming, generating random numbers uniformly within a range is crucial.
- **Statistical Sampling:** In statistical sampling and surveying, discrete uniform distributions can model scenarios where each member of a population has an equal chance of being selected.
- **Modeling Equally Likely Events:** Situations where each outcome has an equal probability, such as selecting a card from a standard deck can be modeled using a discrete uniform distribution.

c) Where can we see it in real life?

- i. **Dice Rolling:** This involves rolling fair dice, such as six-sided or other-sided dice, with an equal chance of every face appearing.



- ii. **Raffles and Lotteries:** selecting tickets or numbers at random from a limited range, with each ticket or number having an equal chance of being chosen. Lotteries, raffles, and random prize drawings are examples of this.



- iii. **Sampling and Selection:** Choosing objects at random, impartially from a limited set so that every item has an equal chance of being selected.
- iv. **Randomized Experiments:** In scientific research, people are assigned at random, without regard to preference, to treatments or conditions. Experiments are fair because every subject has an equal chance of receiving a particular treatment.
- v. **Randomized Algorithms:** Randomization is used by several algorithms to achieve a variety of goals, such as load balancing, pivot point selection, and random shuffling. In these situations, a uniform discrete distribution aids in evenly producing random indices or values.

- vi. **Digital Encryption and Cryptography:** To improve security in cryptographic systems, randomly generate keys, initialization vectors, or nonces in a uniform manner. Secure key creation is influenced by the idea of equally likely possibilities within a certain range.

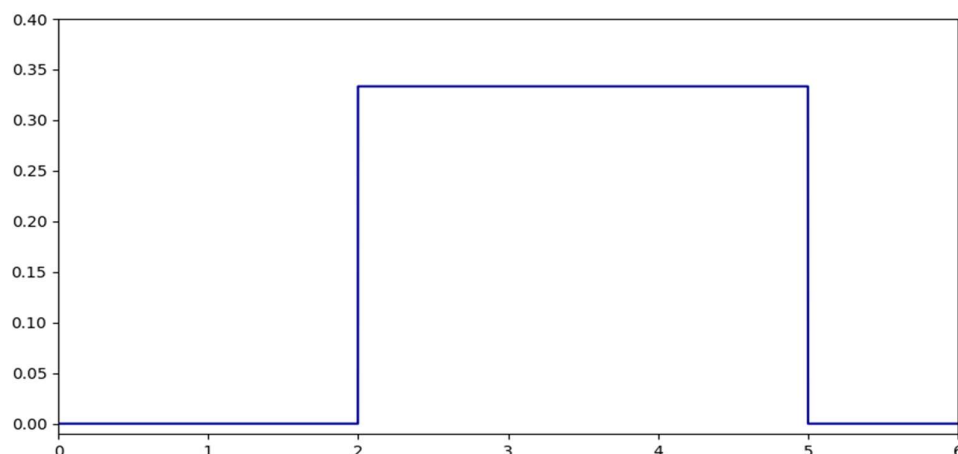


Continuous Random Variables

1. Uniform Continuous distribution

The continuous uniform distributions, sometimes known as rectangle distributions in probability theory and statistics, are a family of symmetric probability distributions. A distribution like this characterizes an experiment in which the result is arbitrary and falls between specific bounds. The parameters, a and b , which represent the minimum and maximum values, define the limits. The interval may be open (i.e. $]a,b[$) or closed (i.e. $[a,b]$). Consequently, the distribution is frequently represented by the acronym, $U(a,b)$, where U denotes a uniform distribution. Interval length is defined as the difference between the boundaries; on the support of the distribution, all intervals of the same length are equally likely. With no extra restrictions, it is the highest entropy probability distribution for a random variable X included in the distribution's.

The following graph shows the distribution with $a = 2$ and $b = 5$:



a) some of its properties

- **Probability density function**

The probability density function of the continuous uniform distribution is:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b. \end{cases}$$

When a=minimum value and b=maximum value

- **Cumulative distribution function**

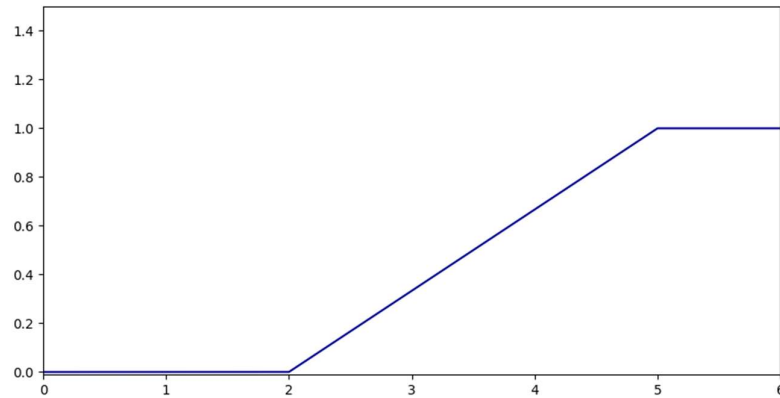
The CDF of a uniform random variable can be determined by integrating $f(x)$:

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x f_X(t) dt \\ &= \int_a^x \frac{1}{b-a} dt \\ &= \frac{x-a}{b-a}, \quad a \leq x \leq b. \end{aligned}$$

Therefore, the complete CDF is:

$$F(x) = \begin{cases} 0 & \text{for } x < a, \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b, \\ 1 & \text{for } x > b. \end{cases}$$

The following graph shows CDF with $a = 2$ and $b = 5$:



- **Variables: -**

The expected value (i.e. the mean or first moment) of a uniform random variable X is:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_a^b \frac{x}{b-a} dx = \frac{a+b}{2},$$

The second moment of a uniform random variable is:

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2 + ab + b^2}{3},$$

The variance of a uniform random variable is:

$$\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{(b-a)^2}{12}.$$

b) Applications: -

- **Randomized Controlled Trials:** In medical research, uniform distributions are used to randomly assign participants to different treatment groups. By ensuring equal probability of assignment, the researchers can minimize bias and create more reliable results.
- **Simulations and Gaming:** Uniform distributions are commonly used in computer simulations and gaming to generate random numbers. They are used, for example, to simulate dice rolls or generate random coordinates on a game map.
- **Quality Control:** Uniform distributions are utilized in quality control processes to randomly sample items from a production line. This helps ensure that the sampling process is unbiased and representative of the entire batch.
- **Random Number Generation:** Uniform distributions play a crucial role in generating random numbers for various applications. They are used in cryptography, Monte Carlo simulations, statistical modeling, and many other areas where randomness is required.
- **Load balancing:** Uniform distributions can be utilized for load balancing in distributed systems and computer networks. For instance, it is possible to divide incoming requests equally among several servers, guaranteeing a more equitable and effective distribution of resources.
- **Lottery and Gambling:** In lotteries and gambling games, uniform distributions are frequently used. They guarantee that every potential result has an equal probability of happening, resulting in an impartial and equitable system for choosing winners.
- **Sports Scheduling:** Uniform distributions can be used to arrange teams in a way that guarantees fairness and equal opportunity for all players when making schedules for leagues or tournaments.
- **Queuing Theory:** The uniform distribution can be used to simulate the arrival times of patrons or events in queuing systems. It aids in the analysis and improvement of the functionality of systems such as contact centers, transportation networks, and service centers.

2. Exponential Continuous Distribution

a) Why was the exponential distribution necessary to create?

- to estimate how long it will take for the next event—a success, failure, arrival, etc.—to occur.
- For instance, the following is what we hope to predict:
The duration of time it takes a customer to complete their browsing and make a purchase from your store (success).

b) What is the origin of exponential random variables?

- An exponential random variable is the inter-arrival time between two consecutive Poisson events. That is, how much time it takes to go from N Poisson counts to $N + 1$ Poisson counts.
- The inter-arrival time T between two consecutive Poisson events is an exponential random variable.

c) Definitions

- PDF

Let X be an exponential random variable. The PDF of X is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$ is a parameter. We write

$$X \sim \text{Exponential}(\lambda)$$

to say that X is drawn from an exponential distribution of parameter λ .

- CDF

CDF

The CDF of an exponential random variables can be determined by

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x f_X(t) dt \\ &= \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}, \quad x \geq 0. \end{aligned}$$

Therefore, if we consider the entire real line, then the CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

- Draw form of (PDF) & (CDF)

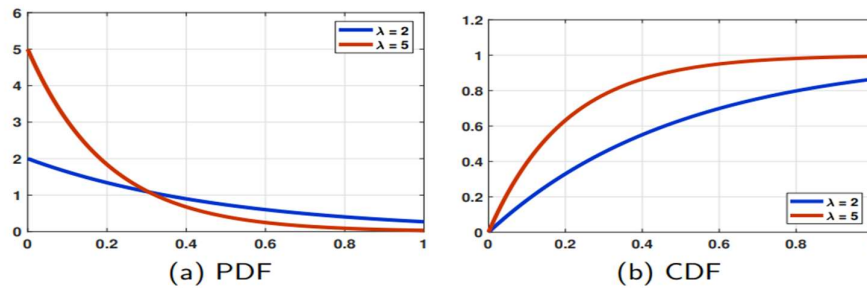


Figure: The PDF and CDF of $X \sim \text{Exponential}(\lambda)$.

Note that the initial value $f_X(0)$ is

$$f_X(0) = \lambda e^{-\lambda \cdot 0} = \lambda.$$

- $E[X]$ & Var

If $X \sim \text{Exponential}(\lambda)$,

then $E[X] = \frac{1}{\lambda}$, and $\text{Var}[X] = \frac{1}{\lambda^2}$

d) Applications

- A waiting time has an exponential distribution if the probability that the event occurs during a certain time interval is proportional to the length of that time interval.

More precisely, has an exponential distribution if the conditional probability.

$$P(t < X \leq t + \Delta t, X > t)$$

is approximately proportional to the length Δt of the time interval comprised between the times t and $t + \Delta t$, for any time instant.

In several practical situations this property is realistic. This is the reason why the exponential distribution can be used to model waiting times.

- Memoryless property

One of the most important properties of the exponential distribution is the memoryless property:

$$P(X \leq x+y, X > x) = P(X \leq y)$$

Proof

X is the time we need to wait before a certain event occurs. The above property says that the probability that the event happens during a time interval of length y is independent of how much time has already elapsed (x) without the event happening.

3. Gaussian Continuous Distribution

a) What is Gaussian distribution?

The Gaussian Random Variable is considered to be the most important continuous random variable due to the fact that it is widely used in almost all scientific disciplines.

Gaussian random variables have two parameters (μ , σ^2). It is noteworthy that the mean is μ and the variance is σ^2 , these two parameters are exactly the first moment and the second central moment of the random variable. Most other random variables do not have this property.

Because the normal distribution is a continuous distribution, we cannot calculate exact probability for an outcome, but instead we calculate a probability for a range of outcomes (for example the probability that a random variable X is greater than 10).

The PDF of the Gaussian Random Variable is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

b) the Graph representation of the PDF:

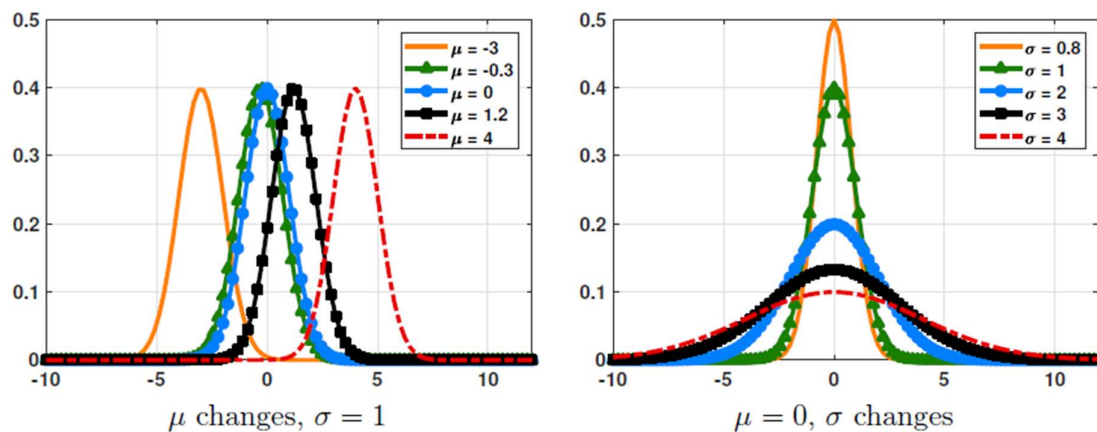


Fig. A Gaussian random variable with different μ and σ .

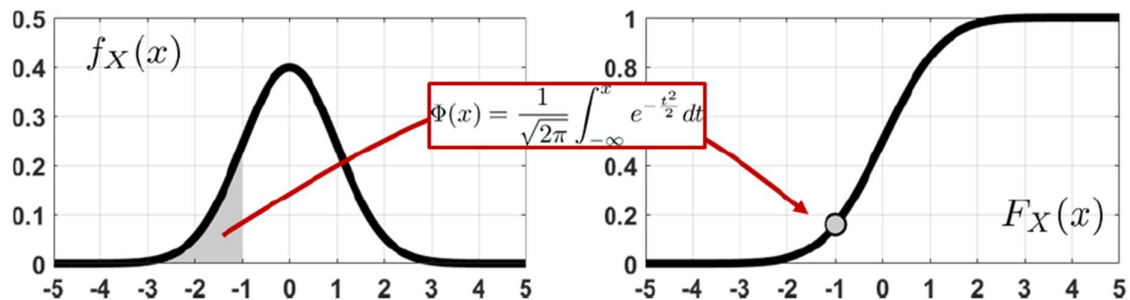
As it is shown in the upper figures, the graphs take the shape of a bell; the distribution is often called the bell curve because of that. In its nature the Gaussian distribution is called normal because it is how most normal data will look when plotted, like measuring the distribution of the height across all high schools for example, or the distribution of the grades of a university. So, you will find that the majority is concentrated in the middle, the average, and the rest are one the other two ends of the graph. Those graphs also demonstrate how the curve will change with the changing of μ or σ .

c) The CDF of the Gaussian distribution:

The CDF of the standard Gaussian is defined as the $\Phi()$ function:

$$\Phi(x) = F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

As for the Graph representation of the CDF:



In the end, we can define the CDF of an arbitrary Gaussian as:

$$F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

Technical discussion:

1. Discrete Bernoulli Implementation:

Import necessary libraries: The code imports the required libraries, including numpy for numerical operations, matplotlib.pyplot for plotting and creating visualizations, and scipy.stats for Bernoulli Distribution.

Then we must define the probability of success for medical treatment. It is set to 0.8 in this example, signifying an 80% chance of success.

Number of patients, '*num_patients*,' defines the number of patients in the simulated dataset. For this illustration, it is set to 1000.

Afterwards, we have to generate synthetic data. '*bernoulli.rvs(p_success, size=num_patients)*,' generates synthetic data for treatment outcomes. It simulates outcomes for '*num_patients*' patients based on the specified probability of success.

`np.sum(treatment_outcomes == 1)` counts the number of patients with a treatment outcome of 1 (success).

`np.sum(treatment_outcomes == 0)` counts the number of patients with a treatment outcome of 0 (failure).

Visualization: A bar plot is created to visualize the distribution of treatment outcomes. The x-axis represents outcomes ('Success' and 'Failure'), and the y-axis represents the counts. The plot is displayed for a quick visual understanding.

2. Discrete Binomial Implementation:

This code uses the `binom` function from the `scipy.stats` library to create a binomial random variable after importing the required libraries. The chance of success in each trial (p) and the number of trials (n) define the binomial random variable.

The number of trials and the value of the random variable are then printed by the code, which then creates a random variable using `rv.rvs()`. The binomial random variable is assigned a random value via the `rv.rvs()` function.

Next, the code generates a Probability Mass Function (PMF) plot for the binomial random variable using the `pmf` method. The PMF plot shows the probability of each possible outcome of the random variable. The x-axis represents the number of successes, and the y-axis represents the probability. The plot is generated using the `plot` function from `matplotlib.pyplot` and labeled with the x and y axis labels and a title.

Next, using the `cdf` technique, the code creates a Cumulative Distribution Function (CDF) graphic for the binomial random variable. The cumulative probability of all possible outcomes up to a specific moment is displayed on the CDF plot. The probability is shown on the y-axis, and the number of successes is represented on the x-axis. The `plot` function from `matplotlib.pyplot` is used to create the plot, which is then labeled with a title and the labels for the x and y axes.

Lastly, the code uses the `mean` and `var` methods to compute and output the variance and expectation of the binomial random variable, respectively. The variance is a measurement of the random variable's dispersion, and the expectation (mean) is the random variable's average value.

3. Discrete Geometry Implementation

The code imports necessary libraries: **numpy** for numerical operations, **matplotlib.pyplot** for plotting, and **scipy.stats.geom** for the geometric distribution.

The **geometric_random_variables** function generates random variables from a geometric distribution with a specified probability of success (**prob_success**) and size (**size**).

```
*if prob_success <= 0 or prob_success > 1:raise ValueError("Probability of success must be in the range (0, 1].")*
```

The function checks if the probability of success is within a valid range (exclusive of 0 and inclusive of 1). If not, a **ValueError** is raised.

```
*geometric_values = np.random.geometric(p=prob_success, size=size)*
```

Random variables are generated using **numpy's random.geometric** function.

This function, **geometric_pmf**, calculates the Probability Mass Function (PMF) of a geometric distribution given the probability of success (**p**) and the number of trials (**k**). It returns a dictionary where keys are the number of trials, and values are the corresponding probabilities.

The **geometric_cdf** function calculates the Cumulative Distribution Function (CDF) based on the previously defined PMF. It returns a dictionary representing the CDF.

This function, **calculate_statistics**, takes a PMF dictionary and calculates the expected value ($E(X)$), expected value of X^2 ($E(X^2)$), and variance ($Var(X)$).

The **plot_geometric_distribution** function generates random variables, calculates the PMF and CDF, and plots the results.

The probability mass function (PMF) and cumulative distribution function (CDF) are calculated using **scipy.stats.geom**.

The PMF is plotted using **plt.stem**, creating blue stems, lines, and markers for visual representation and the CDF is plotted as a step plot using **plt.step**.

This line calculates the expected value ($E(X)$), expected value of X^2 ($E(X^2)$), and variance ($Var(X)$) using the calculated PMF.

Then print the results.

4. Discrete Poisson Implementation:

This code uses the Poisson module from SciPy. Stats library to generate and plot the probability mass function (PMF) and cumulative distribution function (CDF) of a Poisson distribution with a mean of 5.

The code starts by importing the necessary libraries: matplotlib.pyplot for plotting, NumPy for numerical computing, and SciPy. Stats for generating random samples from the Poisson distribution.

We started to put the range of number of orders arriving per day in **np.arange** method we choose a range from 0 to 10.

rv = stats.Poisson: we declared a random variable as a Poisson random variable and it takes one parameter which is lambda.

f = rv.pmf(x): we find the probability mass function of number of orders arriving per day and we put this PMF in variable (f).

r = np.cumsum(f) we summed PMFs to find the cumulative distribution function (CDF).

5. Discrete Uniform Implementation:

The uniform discrete random variable X is created using this code, and its cumulative distribution function (PDF) and probability mass function (PMF) are plotted. Additionally, it calculates X 's expectation and variance.

The variables **a** and **b**, which have values 1 and 13, respectively, determine the range of the random variable X . Thus, X can have values of 1, 2, 3, 4, and up to 13.

The code makes use of the NumPy library's **np.random.randint** function to create the random variable X . Using a uniform distribution throughout a specified range, this function produces a random integer. The number of random integers to be generated is specified by the size argument. Here, the function creates 10,000 random integers from the range [1, 13] since the size is set to 10,000.

Next, the code uses the **plt.hist** function from the Matplotlib library to plot the PMF of X . The range of values that the histogram should cover is specified by the **bins** option. When the **density** argument is set to **True**, the histogram is normalized and the area under the curve equals 1. Plotting the resultant histogram involves connecting a series of "o" markers with lines. Plot labels are added using the **plt.xlabel**, **plt.ylabel**, and **plt.title** methods. The plot is finally displayed by the **plt.show** function.

The CDF of X is then plotted by the code using the **plt.hist** function with the **cumulative** argument set to **True** in a similar fashion. As a result, the function plots the cumulative total of the values in the histogram, producing a monotonically increasing curve that begins at 0.

Lastly, the code uses the formulas for a uniform discrete random variable to calculate the expectation and variance of X . The expectation is just the range's average of its lowest and greatest values, or $(1 + 13) / 2 = 7$ in this instance. The formula $((b - a + 1) ** 2 - 1) / 12$ yields the variance; in this example, it is $((13 - 1 + 1) ** 2 - 1) / 12 = 168 / 12 = 14$.

Using the **print** function, the code outputs the expectation and variance values.

6. Continuous Uniform Implementation:

Import necessary libraries: The code imports the required libraries, including numpy for numerical operations, matplotlib.pyplot for plotting and creating visualizations, and scipy.stats for Uniform Distribution.

This code generates a set of 1000 random samples from a continuous uniform distribution with lower bound and upper bound ask user, plots the probability density function (PDF) and cumulative density function (CDF) of the samples, and calculates the expectation and variance of the uniform distribution.

`stats.uniform.rvs` this is function generates the random value.

`stats.uniform.pdf` this is function make pdf.

`stats.uniform.cdf` this is function make cdf.

`plt.show()` this is function using to display.

Finally, the expectation of the uniform distribution is calculated as the average of the lower and upper bounds, and the variance is calculated using the formula for the variance of a continuous uniform distribution. The results are printed on the console.

7. Continuous Exponential Implementation

This code generates and plots a probability density function (PDF) and cumulative distribution function (CDF) for a set of exponential random variables. It begins by importing the matplotlib.pyplot and numpy libraries, which are used for plotting and generating random numbers, respectively. The pyplot module is imported as plt and the numpy module is imported as np.

```
np.random.exponential(1 / lambda_, n)
```

np.random.exponential(scale, size): This function generates random numbers from an exponential distribution. scale is set to $1 / \lambda$, where λ is the rate parameter of the exponential distribution.

size or n specifies the number of samples to be generated from the exponential distribution.

The range from $x_{\min}$ to $x_{\max}$ is evenly divided into 50 intervals by the np.linspace() function.

The PDF is calculated using the np.histogram() function, which counts the data points in each bin and normalizes the result by the sum of the data points and the width of each bin.

np.cumsum you're essentially summing up the areas of each histogram bin (calculated from the PDF) and accumulating these values to generate the cumulative distribution function (CDF).

Using the **print** function, the code outputs the expectation and variance values.

8. Continuous Gaussian Implementation:

Import necessary libraries: The code imports the required libraries, including numpy for numerical operations, matplotlib.pyplot for plotting, and scipy.stats.norm for the Gaussian distribution.

Define the mean and standard deviation: The mean is set to 0, and the standard deviation is set to 1 for the Gaussian distribution.

Generate random samples: The code uses `np.random.normal` to generate 10,000 random samples from the Gaussian distribution with the specified mean and standard deviation. These samples are stored in the 'samples' variable.

Plot the PMF (Probability Mass Function): The code creates a histogram of the samples using `plt.hist`. The '`density=True`' argument normalizes the histogram to represent probabilities. The resulting PMF is displayed using `plt.show()`.

Plot the PDF (Probability Density Function): The code generates a set of x-values ranging from -5 to 5 using `np.linspace`. It then calculates the corresponding PDF values for the Gaussian distribution using `norm.pdf`. The PDF curve is plotted using `plt.plot` and displayed using `plt.show()`.

Plot the CDF (Cumulative Distribution Function): The code creates a histogram of the samples with the '`cumulative=True`' argument, which computes the cumulative probabilities. The resulting CDF is displayed using `plt.show()`.

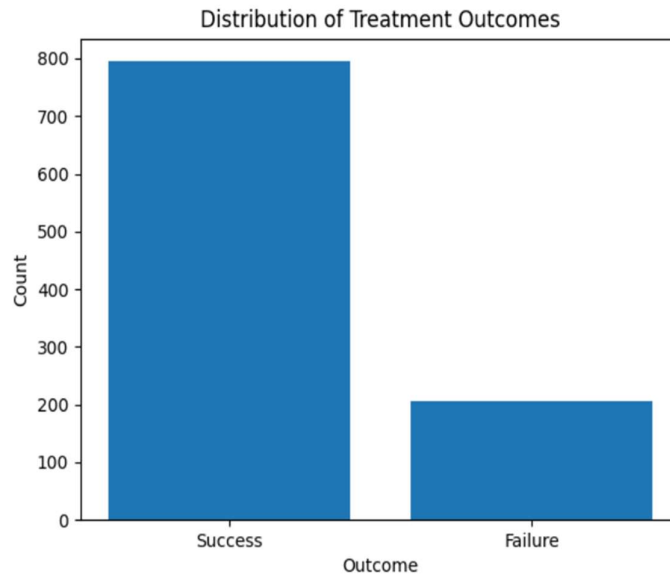
Calculate the expectation: The code calculates the expectation (mean) of the samples using `np.mean` and assigns it to the 'expectation' variable.

Calculate the variance: The code calculates the variance of the samples using `np.var` and assigns it to the 'variance' variable.

Print the expectation and variance: The code prints the calculated expectation and variance using the f-string notation.

Results:

1. Discrete Bernoulli:

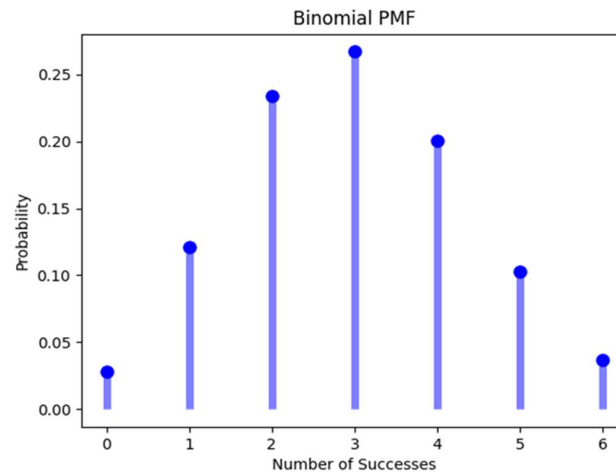


- Expectation & variance

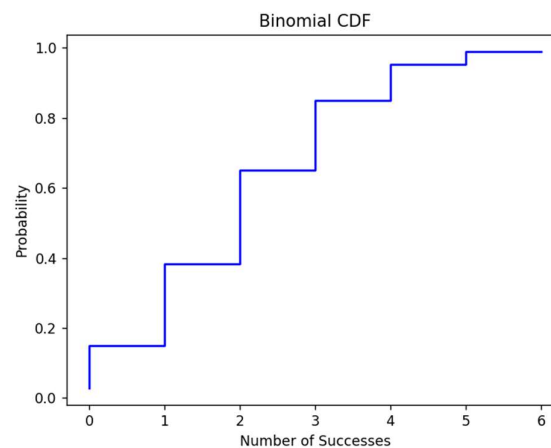
```
Success Count: 788  
Failure Count: 212  
Expectation (Mean): 0.8  
Variance: 0.15999999999999998
```

2. Discrete Binomial:

- PMF



- CDF

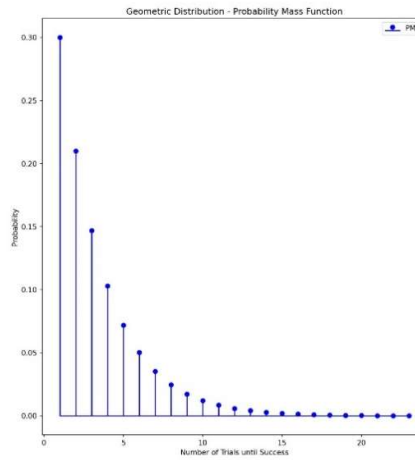


- Expectation & variance

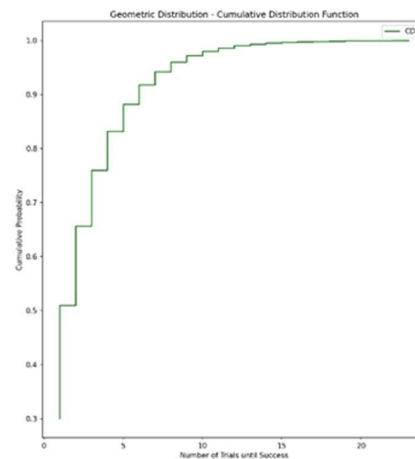
```
Random variable value: 2  
Expectation: 3.0  
Variance: 2.0000000000000005
```

3. Discrete Geometry:

- PMF



- CDF



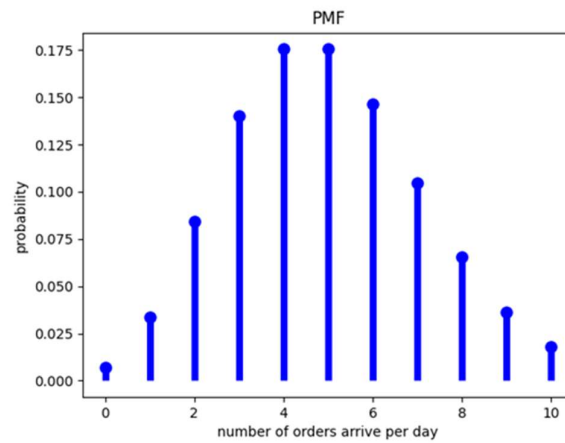
- Expectation & variance

```
Enter the probability of success (0 < prob_success <= 1): 0.3  
Enter the number of random variables to generate: 1000
```

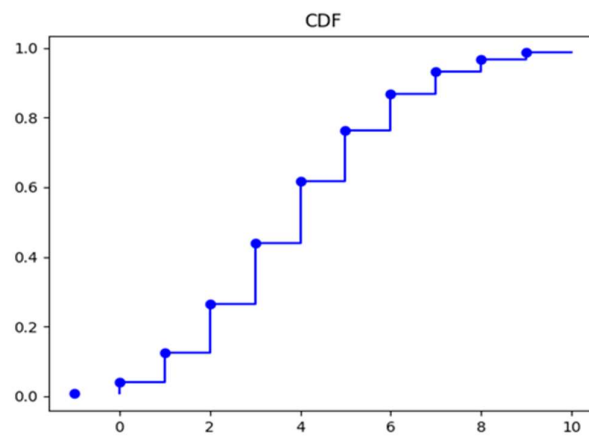
```
Expected Value (E(X)): 2.96  
Expected Value of X^2 (E(X^2)): 13.65  
Variance (Var(X)): 4.91  
|
```

4. Discrete Poisson:

- PMF



- CDF

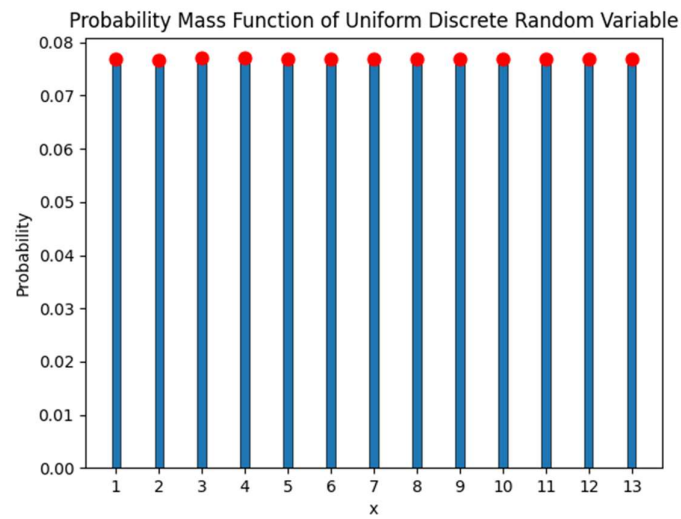


- Expectation & variance

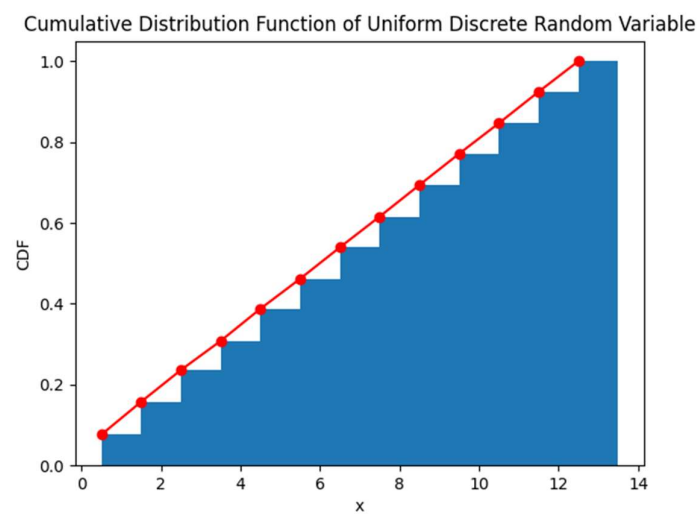
```
expectation = 5  
variance = 5
```

5. Discrete Uniform:

- PMF



- CDF

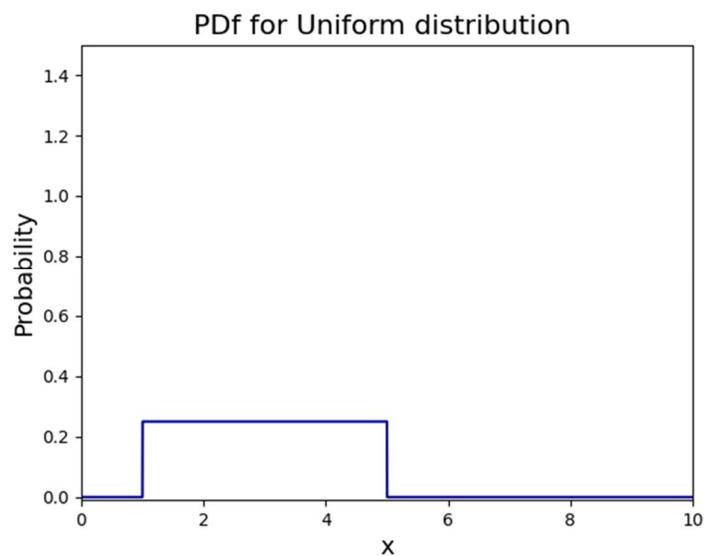


- Expectation & variance

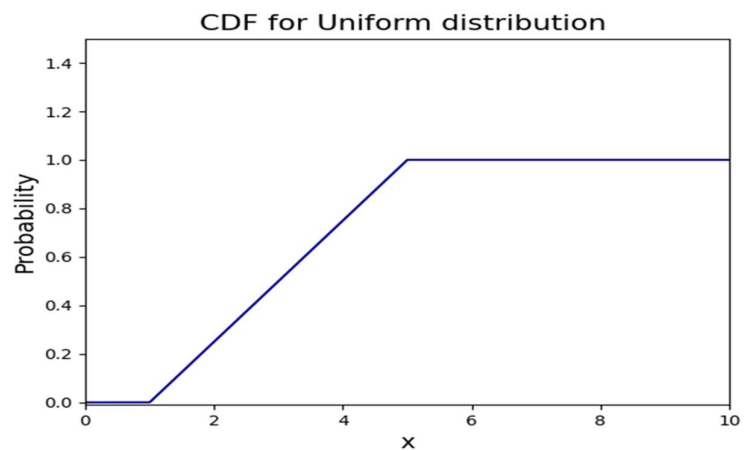
```
Expectation: 7.0  
Variance: 14.0
```

6. Continuous Uniform:

- PDF



- CDF

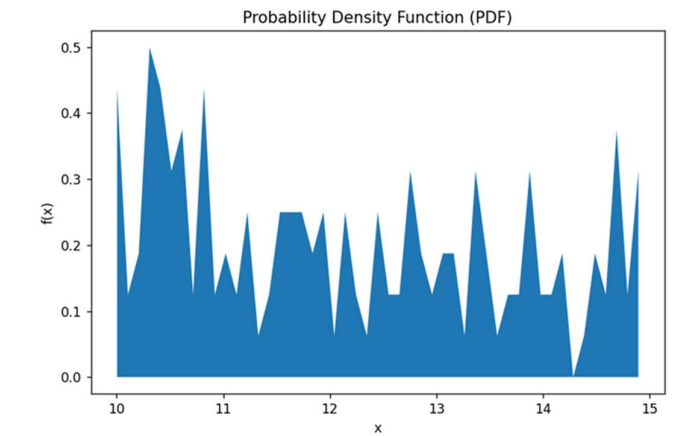


- Expectation & variance

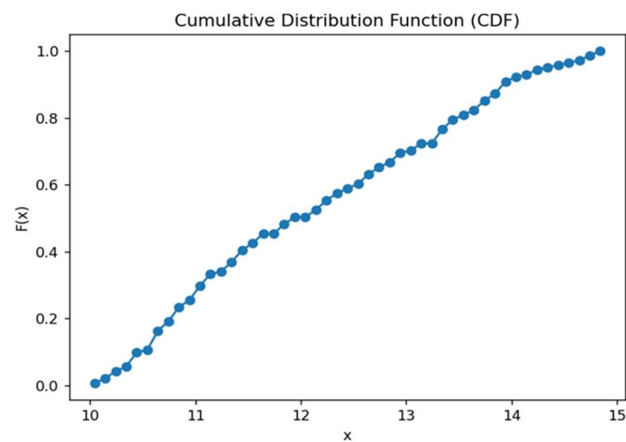
```
Enter minimum value: 1
Enter maximum value: 5
Mean= 3.034090198452379
variance= 1.2911707909299683
```

7. Continuous Exponential:

- PDF



- CDF

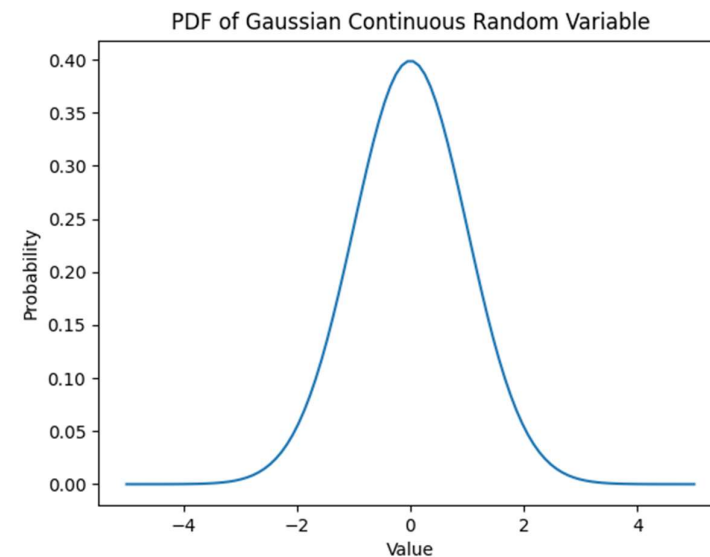


- Expectation & variance

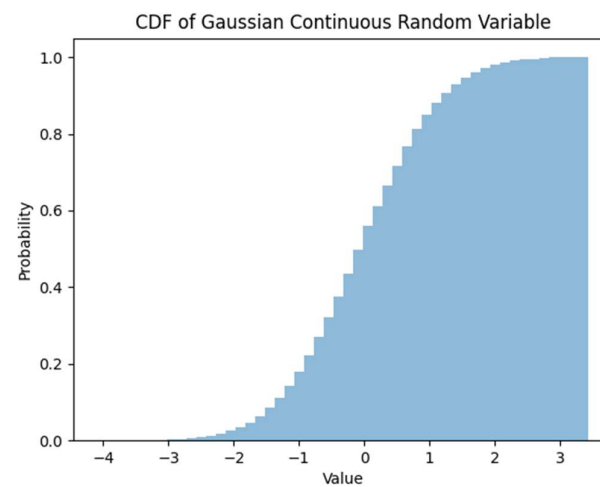
```
Expectation: 10.00  
Variance: 100.00
```

8. Continuous Gaussian:

- PMF



- CDF



- Expectation & variance

```
Expectation: -0.023781855807263986  
Variance: 1.0363799537903327
```

Real life examples:

1. Real-Life Example of the Exponential Distribution:

- Time between calls (applied in the code)

The time between customer calls at different businesses can be modeled using an exponential distribution.

For example, suppose a bank receives a new call every 10 minutes, on average. After a customer calls, find the probability that a new customer calls within 10 to 15 minutes.

To solve this, we start by knowing that the average time between calls is 10 minutes. Thus, the rate can be calculated as:

- $\lambda = 1/\mu$
- $\lambda = 1/10$
- $\lambda = 0.1$

We can use the following formula to calculate the probability that a new customer calls within 10 to 15 minutes:

- $P(10 < X \leq 15) = (1 - e^{-0.1(15)}) - (1 - e^{-0.1(10)})$
- $P(10 < X \leq 15) = .7769 - .6321$
- $P(10 < X \leq 15) = 0.1448$

2. Real-Life Application on Gaussian distribution:

One of the Applications of the distribution is in the field of image processing, specifically in noise modeling and restoration.

In image processing, noise is often introduced during image acquisition, transmission, or storage, resulting in degraded image quality. The Gaussian distribution is frequently used to model additive noise, such as Gaussian noise, which is characterized by random fluctuations that follow a Gaussian distribution.

Assuming that the noise follows a Gaussian distribution, various image restoration techniques can be applied to remove or reduce the noise and restore the original image.

Here's an example of how the Gaussian distribution can be used to model and remove Gaussian noise from an image in Python using the OpenCV library:

```

import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load the noisy image
image = cv2.imread('filename: 'noisy_image.png', flags: 0) # Grayscale image

# Add Gaussian noise to the image
mean = 0
std_dev = 50
noise = np.random.normal(mean, std_dev, image.shape).astype(np.uint8)
noisy_image = cv2.add(image, noise)

# Apply Gaussian blur to denoise the image
denoised_image = cv2.GaussianBlur(noisy_image, ksize: (5, 5), sigmaX: 0)

# Display the original, noisy, and denoised images
plt.figure(figsize=(10, 8))
plt.subplot(131), plt.imshow(image, cmap='gray')
plt.title('Original Image'), plt.axis('off')
plt.subplot(132), plt.imshow(noisy_image, cmap='gray')
plt.title('Noisy Image'), plt.axis('off')
plt.subplot(133), plt.imshow(denoised_image, cmap='gray')
plt.title('Denoised Image'), plt.axis('off')
plt.tight_layout()
plt.show()

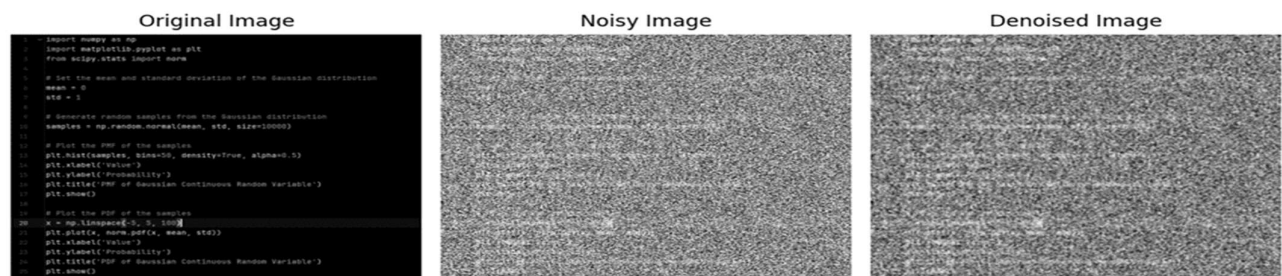
```

In this code, we start by loading a noisy image using `cv2.imread()` function from OpenCV. We then add Gaussian noise to the image by generating random values from a Gaussian distribution with a mean of 0 and a standard deviation of 50 using `np.random.normal()`. The noise is added to the original image using `cv2.add()`.

To denoise the image, we apply a Gaussian blur using `cv2.GaussianBlur()`. This operation convolves the image with a Gaussian kernel, effectively smoothing out the noise while preserving the underlying image structure.

Finally, we display the original, noisy, and denoised images using `plt.imshow()` and `plt.subplot()`. The resulting plot showcases the effect of denoising using the Gaussian distribution for modeling and removing Gaussian noise from the image.

Here is the code's Output using an Image of the code itself!!



3. Real-Life Application on Binomial Distribution:

Banks use binomial distribution to model the probability that a certain number of credit card transactions are fraudulent.

For example, suppose it is known that 2% of all credit card transactions in a certain region are fraudulent. If there are 50 transactions per day in a certain region, we can use a Binomial Distribution Calculator to find the probability that more than a certain number of fraudulent transactions occur in a given day:

- $P(X > 1 \text{ fraudulent transaction}) = \mathbf{0.26423}$
- $P(X > 2 \text{ fraudulent transactions}) = \mathbf{0.07843}$
- $P(X > 3 \text{ fraudulent transactions}) = \mathbf{0.01776}$

And so on.

This gives banks an idea of how likely it is that more than a certain number of fraudulent transactions will occur in a given day.

4. Real-Life Application on Poisson Distribution:

Restaurants use the Poisson distribution to model the number of expected customers that will arrive at the restaurant per day.

For example, suppose a given restaurant receives an average of 100 customers per day. We can use the Poisson distribution calculator to find the probability that the restaurant receives more than a certain number of customers:

- $P(X > 110 \text{ customers}) = \mathbf{0.14714}$
- $P(X > 120 \text{ customers}) = \mathbf{0.02267}$
- $P(X > 130 \text{ customers}) = \mathbf{0.00171}$

And so on.

This gives restaurant managers an idea of the likelihood that they'll receive more than a certain number of customers in a given day.

task assignment

Abdulahman Ghareeb AbdulSattar Muhammad	• Continuous Uniform code • 17% of Report • 11% of Presentation Total: 14%
Ahmed Hamdi AbdulSattar Jadallah	• Continuous Exponential code • 11% of Report • 11% of Presentation Total: 11%
Hisham Ahmed Mohamed Abaza	• Discrete Uniform code • 17% of Report • 11% of Presentation Total: 14%
Abdulaziz Ahmed Abdulaziz Abdulnabi	• Discrete Bernoulli code • 11% of Report • 17% of Presentation Total: 14%
Muhannad Mahmoud AbdulWahab Ahmed	• Discrete Geometric code • 11% of Report • 17% of Presentation Total: 14%
Ahmed Amr Ahmed Hashem	• Continuous Gaussian code • 11% of Report • 11% of Presentation Total: 11%
Rawan Abdulaziz Ali Abdulaziz	• Discrete Poisson code • 11% of Report • 11% of Presentation Total: 11%
Ahmed Mahmoud Mohamed Abdullatif	• Discrete Binomial code • 11% of Report • 11% of Presentation Total: 11%

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